Name : Nisha Yuvraj Pragane | DIV:SEDA | Roll no :13 | PRN : 1252090013 | BATCH :B1 Write a program to do following: This dataset gives the data of Income and money spent by the customers visiting a shopping mall. The data set contains Customer ID, Gender, Age, Annual Income, Spending Score. Therefore, as a mall owner you need to find the group of people who are the profitable customers for the mall owner. Apply at least two clustering algorithms (based on Spending Score) to find the group of customers. Apply Data pre-processing Perform data-preparation (Train-Test Split) Apply Machine Learning Algorithm Evaluate Model. Data Set: https://www.kaggle.com/shwetabh123/mall-customers

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.model_selection import train_test_split
from sklearn.metrics import silhouette_score
```

Load Dataset

```
df = pd.read_csv('Mall_Customers.csv')
df.head()
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Genre                   200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.drop('CustomerID',axis=1,inplace=True)
df
```

	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	Male	19	15	39
1	Male	21	15	81
2	Female	20	16	6
3	Female	23	16	77
4	Female	31	17	40
...	...	...	...	...
195	Female	35	120	79
196	Female	45	126	28
197	Male	32	126	74
198	Male	32	137	18
199	Male	30	137	83

200 rows × 4 columns

Label encoding

```
df['Genre'] = df['Genre'].map({'Male': 0, 'Female': 1})
df
```

	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	0	19	15	39
1	0	21	15	81
2	1	20	16	6
3	1	23	16	77
4	1	31	17	40
...	...	...	...	...
195	1	35	120	79
196	1	45	126	28
197	0	32	126	74
198	0	32	137	18
199	0	30	137	83

200 rows × 4 columns

Select Features

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
plt.scatter(X.iloc[:,0], X.iloc[:,1])
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Distribution")
plt.show()
```

Feature Scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Implementing train test split in 80/20 ratio

```
X_train, X_test = train_test_split(X_scaled, test_size=0.2, random_state=42)
```

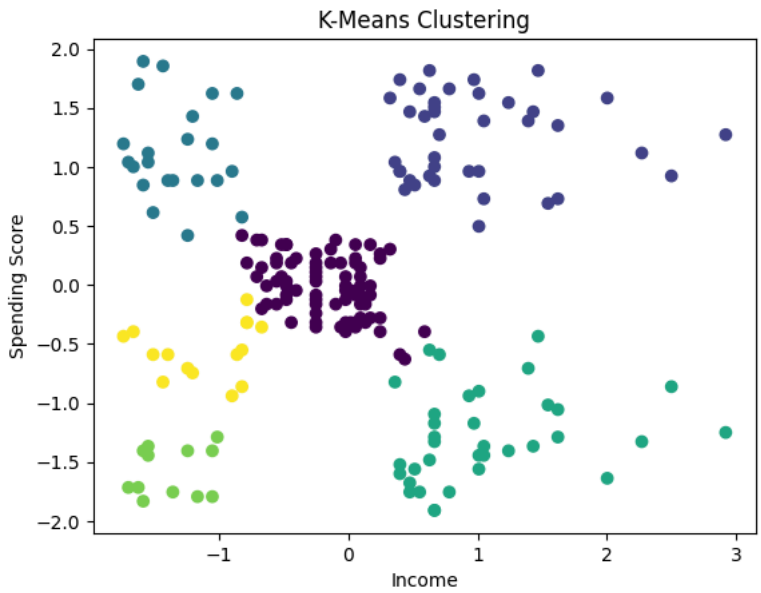
Silhouette Score

```
kmeans_score1 = silhouette_score(X_scaled, y_kmeans)
print("K-Means Silhouette Score:", kmeans_score1)
```

```
K-Means Silhouette Score: 0.5546571631111091
```

K++

```
kmeans3 = KMeans(n_clusters=6, random_state=42)
y_kmeans = kmeans3.fit_predict(X_scaled)
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans, cmap='viridis')
plt.title("K-Means Clustering")
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.show()
```

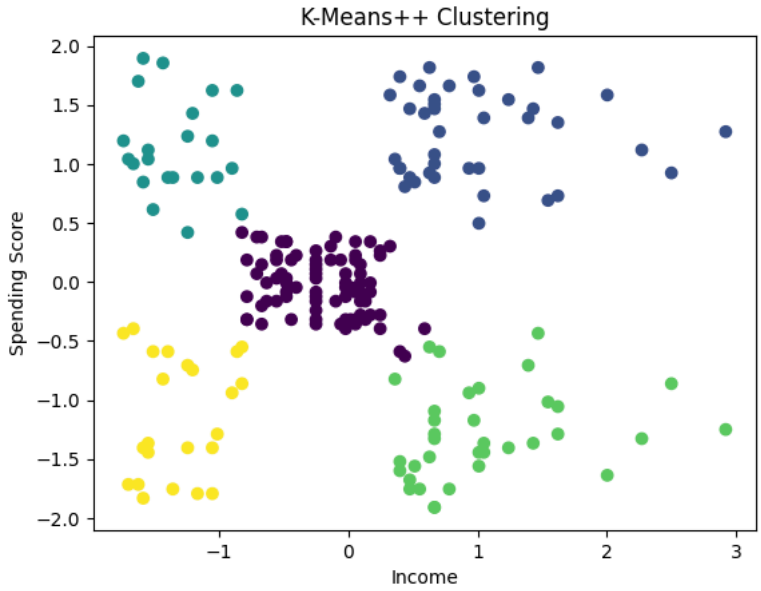


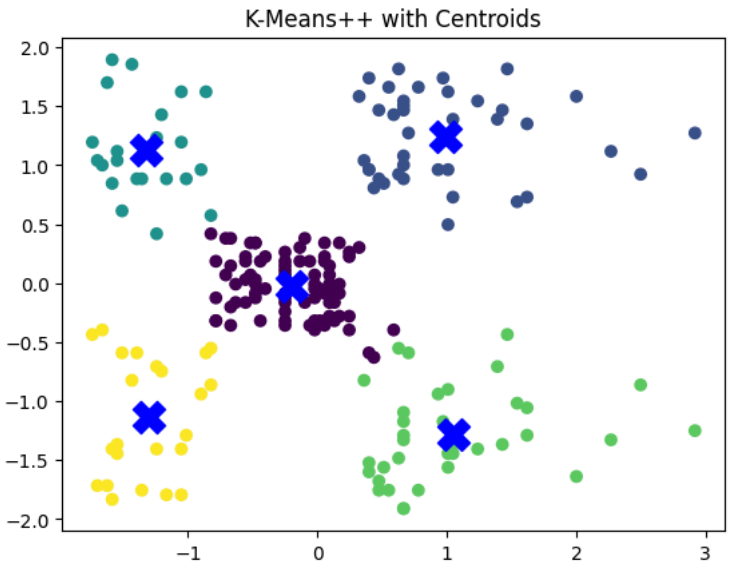
Kmeans++

```
kmeans_pp = KMeans(
    n_clusters=5,
    init='k-means++',
    random_state=42
)

y_kmeans_pp = kmeans_pp.fit_predict(X_scaled)
```

```
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans_pp, cmap='viridis')
plt.title("K-Means++ Clustering")
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.show()
```




```
centroids = kmeans_pp.cluster_centers_

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans_pp, cmap='viridis')
plt.scatter(centroids[:,0], centroids[:,1], s=300, c='blue', marker='X')
plt.title("K-Means++ with Centroids")
plt.show()
```